

# FILTERIZATION

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ABSTRACT. x

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## 1. VF-CATEGORIES

### 1.1. Blass’ category of filters.

**Definition 1.1.** The *category of filters*  $\mathbb{F}$  is defined as follows:

- as objects, we let pairs  $(S, \sigma)$  of a set  $S$  and a filter  $\sigma$  on  $S$ ;
- as morphisms  $(T, \tau) \rightarrow (S, \sigma)$ , we let equivalence classes of functions  $\phi: T \rightarrow S$  such that  $\phi^{-1}(S_0) \in \tau$  for each  $S_0 \in \sigma$ , i.e.  $\sigma \subseteq F\phi(\tau)$  where  $F: \mathbf{Set} \rightarrow \mathbf{Pos}$  is the (relative) filter monad, identified up to equality on a  $\tau$ -large subset.

For ease of notation, we will often omit the set underlying a filter.

Recall that an arrow  $\phi: (T, \tau) \rightarrow (S, \sigma)$  in  $\mathbb{F}$  is monic if and only if it is injective on a  $\tau$ -large subset, in which case it can be identified with the inclusion of a  $\sigma$ -large subset; on the other hand, it is epic if and only if  $\phi(T_0) \in \sigma$  for each  $T_0 \in \tau$ , or equivalently  $\sigma = F\phi(\tau)$ , in which case it can be assumed to be surjective. Moreover,  $\mathbb{F}$  is balanced.

Note also that, for a set  $S$ , the hom-set  $\mathbb{F}(\tau, (S, \{S\}))$  is in bijection with the *reduced power* of  $S$  with respect to  $\tau$ , generalizing ultrapowers to filters.

We note here the following property of  $\mathbb{F}$ , first remarked by the authors of [GMT26].

**Lemma 1.2.** *For any epimorphism  $f: \mu \twoheadrightarrow \nu$  in  $\mathbb{F}$ , there exist a set  $K$ , a filter  $\kappa$  on  $K$ , and an arrow  $g: \kappa \otimes \nu \rightarrow \mu$  such that  $f \circ g$  coincides with the projection  $\pi_J: \kappa \otimes \nu \rightarrow \nu$ :*

$$\begin{array}{ccc}
 & & \mu \\
 & \nearrow g & \downarrow f \\
 \kappa \otimes \nu & \xrightarrow{\pi_J} & \nu
 \end{array}$$

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*Proof.* Up to isomorphism in  $\mathbb{F}$ , we may pick a surjective representative  $f: I \twoheadrightarrow J$  for the arrow  $f: (I, \mu) \rightarrow (J, \nu)$ . Let  $K$  be the set  $\{k: J \rightarrow I \mid fk = \text{id}_J\}$  of sections of  $f$ , which is non-empty as  $f$  is surjective, and define  $g: K \times J \rightarrow I$  by  $g(k, j) := k(j)$ , which clearly gives  $f \circ g = \pi_J$ .

For each  $A \in \mu$ , consider the set  $N_A := \{k \in K \mid (kf)^{-1}(A) \in \mu\}$ . We claim that the family  $\{N_A \mid A \in \mu\}$  has the finite intersection property. As  $N_A \cap N_B = N_{A \cap B}$  for all  $A, B \in \mu$ , it suffices to prove  $N_A \neq \emptyset$  for each  $A \in \mu$ . Indeed, given  $A \in \mu$ , define  $k \in K$  by setting  $k(j)$  to be any element of  $A \cap f^{-1}(j)$  if  $j \in f[A]$ , and an arbitrary element of  $f^{-1}(j)$  otherwise. Then  $A \subseteq (kf)^{-1}(A)$ , so that  $(kf)^{-1}(A) \in \mu$  and hence  $k \in N_A$ . **UT:** this uses classical logic!! (but not choice)

Consider then the filter  $\kappa$  on  $K$  generated by  $\{N_A \mid A \in \mu\}$ . We claim that  $\mu \subseteq g(\kappa \otimes \nu)$ , so that  $g$  defines an arrow  $g: \kappa \otimes \nu \rightarrow \mu$  in  $\mathbb{F}$ . Let  $A \in \mu$ . Note that  $\sum_{k \in N_A} k^{-1}(A) \subseteq g^{-1}(A)$ , by definition of  $N_A$ . Now, for each  $k \in N_A$ , we have  $k^{-1}(A) \in f(\mu) = \nu$ , and by construction  $N_A \in \kappa$ . Thus,  $\sum_{k \in N_A} k^{-1}(A) \in \kappa \otimes \nu$ , so that  $g^{-1}(A) \in \kappa \otimes \nu$  as well, i.e.,  $A \in g(\kappa \otimes \nu)$ .  $\square$

**GS:** What we need to make it work is that  $f: \tau \twoheadrightarrow \sigma$  satisfies that  $\int_{s:\sigma} T^0_s$  is non-empty for all  $T^0 \subseteq T$  that is  $\nu$ -large. Maybe we can replace, in the statement, epi by map satisfying this property?

### Lemma 1.3.

#### 1.2. Vf-categories.

**Definition 1.4.** A *vf-category*  $\mathcal{X}$  is defined by the following data:

- (i) a class  $\mathcal{X}_0$ , whose elements are called *points*;
- (ii) for each  $p \in X_0$  and each  $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$  for some filter  $\sigma \in \mathbb{F}$ , a functor:

$$\mathcal{X}_{p,(q_s)}: (\mathbb{F}/\sigma)^{\text{op}} \rightarrow \mathbf{Set} \quad (\phi: \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t:\tau})$$

where elements of  $\mathcal{X}(p, (q_{\phi t})_{t:\tau})$  are called *vf-arrows* and denoted by  $r: p \multimap_{t:\tau} q_{\phi t}$  or simply  $r: p \multimap_\tau q_{\phi t}$ ;

- (iii) for each  $p, (q_s)_{s:\sigma}, (r_{s,t})_{(s,t):\sum_{s:\sigma} \tau_s}$ , a *unit* and a *composition*:

$$\begin{array}{ccccc} 1 & \longrightarrow & 1 & (\mathbb{F}/\sigma \times \int_{s:\sigma} (\mathbb{F}/\tau_s))^{\text{op}} & \xrightarrow{\mathcal{X}_{p,(q_{-\ell})} \times \int_{s:\sigma} \mathcal{X}_{q_s,(r_{s,-k})}} & \mathbf{Set} \times \mathbf{Set} \\ \star \downarrow & & \Downarrow & \Sigma \downarrow & & \Downarrow \\ \mathbb{F}^{\text{op}} & \xrightarrow{H_{p,(p)}} & \mathbf{Set} & (\mathbb{F}/\sum_{s:\sigma} \tau_s)^{\text{op}} & \xrightarrow{H_{p,-}} & \mathbf{Set} \end{array}$$

This data is required to satisfy the following axioms:

- (i) **axioms**

We say that  $\mathcal{X}$  is *small* if so is  $X_0$ .

For any vf-arrow  $r: p \multimap_{s:\sigma} q_s$  and any  $\phi: \tau \rightarrow \sigma$  in  $\mathbb{F}/\sigma$  — seen as an arrow  $\phi \rightarrow \text{id}_\sigma$  in  $\mathbb{F}/\sigma$  — we denote by  $\phi^*(r)$  the vf-arrow  $\mathcal{X}_{p,(q_s)}(\phi)(r): p \multimap_{t:\tau} q_{\phi t}$ , which we refer to as the *reindexing* of  $r$  along  $\phi$ .

- If  $\phi$  is monic, we also say that  $\phi^*(r)$  is a *specialization* of  $r$ .
- If  $\phi$  is epic, we also say that  $\phi^*(r)$  is a *thickening* of  $r$ .

**1.3. Tautness.** Let  $\mathcal{X}$  be a vf-category. Note first that if a vf-arrow  $u: p \multimap_\tau q_{\phi t}$  is the thickening of  $v: p \multimap_\sigma q_s$  with respect to some  $\mathbb{F}$ -arrow  $\phi: \tau \twoheadrightarrow \sigma$ , i.e.  $u = \phi^*v$ , then in particular  $\psi_1^*u = \psi_2^*v$  for any pair of  $\mathbb{F}$ -arrows  $\psi_1, \psi_2$  such that  $\phi \circ \psi_1 = \phi \circ \psi_2$ .

**Definition 1.5.** A vf-arrow  $u: p \multimap_{\tau} q_{\phi t}$  in  $\mathcal{X}$  is *unthickenable* if  $\psi_1^* u = \psi_2^* u$  for every pair of  $\mathcal{F}$ -arrows  $\psi_1, \psi_2$  such that  $\phi \circ \psi_1 = \phi \circ \psi_2$ .

As we now define, in a *taut* vf-category, unthickenable vf-arrows can be *uniquely unthickened*.

**Definition 1.6.** A vf-category  $\mathcal{X}$  is *taut* if:

- (i) for every unthickenable vf-arrow  $u: p \multimap_{\tau} q_{\phi t}$  there exists a unique vf-arrow  $u_0: p \multimap_{\sigma} q_s$  such that  $u = \phi^* u_0$ ;
- (ii) the canonical map  $\mathcal{X}(p, (q_t)_{t: \coprod_i \sigma_i}) \rightarrow \prod_i \mathcal{X}(p, (q_s)_{s: \sigma_i})$  defined by specialization along each coproduct injection  $\sigma_i \hookrightarrow \coprod_i \sigma_i$  is a bijection.

**Example 1.7.** Tautness can be defined analogously for vu-categories, disregarding the second condition; a vu-poset is taut if and only if it is a topological space. UT: for vf-posets, this is \*stronger\* than the the condition characterizing relational  $F$ -algebras?

**Construction 1.8.** For a vf-poset  $X$ , its *tautification* is the vf-poset  $\rho_T(X)$  defined as follows:

- (i) points of  $\rho_T(X)$  are the same as those of  $X$ ;
- (ii) for each  $p \in X_0$  and  $(q_s)_{s: \sigma} \in X_0^{\sigma}$ ,  $p \preceq_{\sigma} q_s$  holds if and only if there exists a cover  $\phi: \tau \twoheadrightarrow \sigma$  in  $\mathbb{F}$  such that  $p \preceq_{\tau} q_{\phi t}$  holds in  $X$ .

It is evident that  $\rho_T(X)$  is taut, and there is a canonical vf-functor  $\eta_X: X \rightarrow \rho_T(X)$  acting as the identity on points.

We can now give two equivalent characterizations of tautness.

First, recall from [AE04] that we can consider a Grothendieck topology  $J_{\infty}$  on the category  $\mathbb{F}$  of filters, where coverings of  $\sigma \in \mathbb{F}$  are generated by (possibly infinite) jointly-epimorphic families, i.e. families  $\{\gamma_i: \sigma_i \rightarrow \sigma\}_{i \in I}$  such that  $\bigcup_{i \in I} \gamma_i(S_i) \in \sigma$  for any  $\{S_i \in \sigma_i\}_{i \in I}$ . We also denote by  $J_{\infty}$  the topology induced on each slice category of  $\mathbb{F}$ , i.e. where a covering of  $\phi: \tau \rightarrow \sigma$  in  $\mathbb{F}/\sigma$  is a covering of  $\tau$  in  $\mathbb{F}$  seen as a collection of morphisms to  $\phi$  in  $\mathbb{F}/\sigma$ .

**Theorem 1.9.** Let  $\mathcal{X}$  be a vf-category. The following are equivalent:

- (i)  $\mathcal{X}$  is taut;
- (ii) for any small vf-poset  $Y$  and any vf-functor  $F: Y \rightarrow \mathcal{X}$ , there exists an essentially unique lift  $\bar{F}: \rho_T(Y) \rightarrow \mathcal{X}$  of  $F$  along the tautification map  $\eta_Y: Y \rightarrow \rho_T(Y)$ :

$$\begin{array}{ccc} Y & \xrightarrow{F} & \mathcal{X} \\ \eta_Y \downarrow & \nearrow \bar{F} & \\ \rho_T(Y) & & \end{array}$$

- (iii) for any  $p \in \mathcal{X}_0$  and any  $(q_s)_{s: \sigma} \in \mathcal{X}_0^{\sigma}$ , the functor  $\mathcal{X}_{p, (q_s)}: (\mathbb{F}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$  mapping  $(\phi: \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t: \tau})$  is a  $J_{\infty}$ -sheaf on  $\mathbb{F}/\sigma$ .

*Proof.* Towards (1)  $\Rightarrow$  (3), consider a (generating)  $J_{\infty}$ -covering family  $\{\gamma_i: \tau_i \rightarrow \tau\}_{i \in I}$  of  $\phi: \tau \rightarrow \sigma$  in  $\mathbb{F}/\sigma$  and a compatible family  $\{u_i: p \multimap_{\tau_i} q_{\gamma_i \phi t}\}_{i \in I}$  in  $\mathcal{X}_{p, (q_s)}$ . We need to find a unique  $v: p \multimap_{\tau} q_{\phi t}$  such that  $u_i = \gamma_i^* v$  for each  $i \in I$ .

Taking coproducts, we can consider the cover  $\gamma: \coprod_i \tau_i \twoheadrightarrow \tau$ : if we show that  $\gamma^* u$  is unthickenable

- (3)  $\Rightarrow$  (2)
- (2)  $\Rightarrow$  (1)

□

**Remark 1.10.** The Grothendieck topology  $J_{\infty}$  restricts to the atomic topology on  $\mathbb{U}$ . we expect that vu-categories and vf-categories are equivalent notions (see Vu-vF appendix)

#### 1.4. Fully-faithfulness.

**Definition 1.11.** A vf-functor  $F: \mathcal{X} \rightarrow \mathcal{Y}$  is *vf-faithful* (resp. *vf-fully-faithful*) if, for each  $p \in \mathcal{X}_0$  and each  $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$ , the function

$$\mathcal{X}(p, (q_s)_{s:\sigma}) \rightarrow \mathcal{Y}(F(p), (F(q_s))_{s:\sigma})$$

is injective (resp. bijective).

**Definition 1.12.** A vf-functor  $F: \mathcal{X} \rightarrow \mathcal{Y}$  is *vf-full* if for each vf-arrow  $u: F(x) \multimap_{s:\sigma} F(y_s)$  in  $\mathcal{Y}$  there exist a covering  $\{\phi_i: \tau_i \rightarrow \sigma\}_{i \in I}$  of  $\sigma$  in  $\mathbb{F}$  and a family of vf-arrows  $\{v_i: x \multimap_{t:\tau_i} y_{\phi_i t}\}_{i \in I}$  in  $\mathcal{X}$  such that  $F(v_i) = \phi_i^*(u)$  for each  $i \in I$ .

**Remark 1.13.** Taking coproducts in  $\mathbb{F}$ , the previous definition can be reduced to the case  $I = 1$ ; moreover, by Lemma 1.2 we can assume  $\phi = \pi_2: \lambda \cdot \sigma \rightarrow \sigma$ , i.e., there exists an arrow  $g: p \multimap_{\lambda \cdot \sigma} ((q_s)_{s:\sigma})_{\ell:\lambda}$ , with  $\lambda$  inhabited, such that  $F(g) = \pi_2^*(f)$ . **GS: a filter  $\sigma$  is inhabited if  $\sigma \rightarrow \star$  is a covering, classically it means that it is not the initial filter UT: a priori Lemma 1.2 uses classical logic so maybe we avoid it if possible/if we can't make it constructive**

**Lemma 1.14.** Let  $F: \mathcal{X} \rightarrow \mathcal{Y}$  be a vf-functor and suppose that  $\mathcal{X}$  and  $\mathcal{Y}$  are taut. Then, the following are equivalent:

- (i)  $F$  is vf-full and vf-faithful;
- (ii)  $F$  is vf-fully-faithful.

*Proof.* Clearly, (2)  $\Rightarrow$  (1). Conversely, suppose  $F$  is vf-full and vf-faithful. To show that it is vf-fully-faithful we need to show that, for each  $p \in \mathcal{X}_0$  and each  $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$ , the function

$$\mathcal{X}(p, (q_s)_{s:\sigma}) \rightarrow \mathcal{Y}(F(p), (F(q_s))_{s:\sigma})$$

is surjective, so let  $u: F(p) \multimap_\sigma F(q_s)$  be a vf-arrow in  $\mathcal{Y}$ . By vf-fullness of  $F$ , there exist a cover  $\phi: \tau \twoheadrightarrow \sigma$  and a vf-arrow  $v: p \multimap_\tau q_{\phi t}$  in  $\mathcal{X}$  such that  $F(v) = \phi^*u$ . Now, for any  $\psi_1, \psi_2: \lambda \rightarrow \tau$  equalizing  $\phi$ , we have

$$F(\psi_1^*v) = \psi_1^*F(v) = \psi_1^*\phi^*u = \psi_2^*\phi^*u = \psi_2^*F(v) = F(\psi_2^*v)$$

and hence  $\psi_1^*v = \psi_2^*v$  by vf-faithfulness of  $F$ . In other words,  $v$  is unthickenable: since  $\mathcal{X}$  is taut, it follows that there is a (unique) vf-arrow  $v_0: p \multimap_\sigma q_s$  such that  $v = \phi^*v_0$ . Thus,

$$\phi^*u = F(v) = F(\phi^*v_0) = \phi^*F(v_0)$$

and hence  $u = F(v_0)$  by tautness of  $\mathcal{Y}$ .  $\square$

**1.5. Subspaces.** Let  $\mathcal{X}$  be a vf-category. For  $p \in \mathcal{X}_0$  and  $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$ , we write  $p \preceq (q_s)_{s:\sigma}$  if the set  $\mathcal{X}(p, (q_s)_{s:\sigma})$  is inhabited.

**Definition 1.15.** A *subspace*  $\mathcal{Y}$  of  $\mathcal{X}$  is simply a sub-vf-category, i.e. a vf-category defined by restricting the functors  $\mathcal{X}_{p,(q_s)}$  to a subclass  $\mathcal{Y}_0$  of  $\mathcal{X}_0$ , together with the same units and compositions. Clearly, the inclusion  $\mathcal{Y}_0 \hookrightarrow \mathcal{X}_0$  extends to a vf-fully-faithful vf-functor  $\mathcal{Y} \rightarrow \mathcal{X}$ .

A subspace  $\mathcal{Y} \subseteq \mathcal{X}$  is *open* if, for any  $p \in \mathcal{Y}_0$  and any  $(q_s)_{i:\sigma} \in \mathcal{X}_0^\sigma$ , if  $p \preceq (q_s)_{s:\sigma}$  then  $(\forall s: \sigma) q_s \in \mathcal{Y}_0$ . **GS: Open subspaces of  $\mathcal{X}$  correspond exactly to a vf-functor  $\mathcal{X} \rightarrow [2]$  into the Sierpinski vf-category.**

If  $\mathcal{X}$  is small, then the poset of open subspaces of  $\mathcal{X}$  ordered by inclusion is a frame, which we denote by  $\mathcal{O}(\mathcal{X})$ . Note that every vf-functor  $\mathcal{X} \rightarrow \mathcal{Y}$  induces, by precomposition, a frame homomorphism  $\mathcal{O}(\mathcal{Y}) \rightarrow \mathcal{O}(\mathcal{X})$ : this association determines a 2-functor  $\mathbf{vfCat} \rightarrow \mathbf{Loc}$  and hence a 2-functor  $\rho_T: \mathbf{vfCat} \rightarrow \mathbf{TopSp}$ , the *topological reflection*, which is left 2-adjoint to the embedding  $\iota_T: \mathbf{TopSp} \hookrightarrow \mathbf{vfCat}$ .

**1.6. Vf-categories of points.** Let  $\mathcal{E}$  be a topos and fix a full subcategory  $A \subseteq \mathbf{pt}(\mathcal{E})$ , where we identify points of  $\mathcal{E}$  with their inverse images  $\mathcal{E} \rightarrow \mathbf{Set}$ . Generalizing the canonical vu-category structure, we can endow  $A$  with a canonical vf-category structure such that the evaluation map  $\mathbf{ev}: \mathcal{E} \rightarrow [A, \mathbf{Set}]$  factors through  $\mathbf{Sh}(A) := \mathbf{vfCat}(A, \mathbf{Set})$ . *UT: i personally don't like the notation  $\mathbf{Sh}(\mathbf{pt}(\mathcal{E})|_A)$  that much; if anything, i'd just write  $\mathbf{Sh}(A)$  GS: i do agree, i changed it!* Concretely, a vf-arrow  $x \multimap_{s:\sigma} y_s$  in  $A$  is given by a natural transformation  $x \Rightarrow \int_{s:\sigma} y_s$ , where  $\int_{s:\sigma} y_s$  is the *reduced product* functor.

We now show that vu-categories arising as points of a topos are taut.

**Lemma 1.16.** *Let  $\phi: \tau \rightarrow \sigma$  in  $\mathbb{U}$  and  $(A_s)_{s:\sigma}$  a  $\sigma$ -family of sets. Then the image of  $\int_{s:\sigma} A_s \hookrightarrow \int_{t:\tau} A_{\phi t}$  is described by the  $x$  such that  $\psi_1^* x = \psi_2^* x$  for all pair of maps  $\psi_1, \psi_2: \lambda \rightrightarrows \tau$  equalizing  $\phi$ .*

*Proof.* An element  $\int_{t:\tau} A_{\phi t}$  that can be lift to  $\int_{s:\sigma} A_s$  clearly satisfies the condition. We suppose now that  $(a_t) \in \int_{t:\tau} A_{\phi t}$  cannot be lift to  $\int_{s:\sigma} A_s$ . We chose underlying sets  $\phi: T \rightarrow S$  and we claim than the subsets  $T_0 \times_{\phi(T_0)} T_0 \subseteq T \times_S T$ , for  $T_0$   $\tau$ -large, together with the subset  $D = \{(t, t') \in T \times_S T \mid a_t \neq a_{t'}\}$  can be extended to a filter. We can thus extend them into ultrafilter  $\lambda$  on  $T \times_S T$ , the sets  $(T_0 \times_{\phi(T_0)} T_0)$  ensure that the projections give maps  $\psi_1, \psi_2: \lambda \rightrightarrows \tau$  equalizing  $\phi$  and the set  $D$  ensures that  $\psi_1^*(a_t) \neq \psi_2^*(a_t)$ .  $\square$

**Proposition 1.17.** *Any sober vu-category  $\mathcal{X} = \mathbf{pt}(\mathcal{E})$  is taut.*

*Proof.* We take some points  $p, (q_s)_{s:\sigma}$  of  $\mathcal{E}$  and  $f: p \multimap_{\tau} q_{\phi t}$  in  $\mathbf{pt}(\mathcal{E})$ , that is, maps  $f_E: E_p \rightarrow \int_{t:\tau} E_{q_{\phi t}}$  natural in  $E \in \mathcal{E}$ , that can be unthicken along  $\phi: \tau \rightarrow \sigma$ . It is enough to show that for any  $\xi \in E_p$ ,  $f_E(\xi)$  is in the image of  $\int_{s:\sigma} E_{q_s} \hookrightarrow \int_{t:\tau} E_{q_{\phi t}}$ . The hypothesis that  $f$  can be unthicken along  $\phi$  ensures that  $\psi_1^* f_E(\xi) = \psi_2^* f_E(\xi)$  for all  $\psi_1, \psi_2: \lambda \rightrightarrows \tau$  equalizing  $\phi$ , and so we can conclude with Lemma 1.16.  $\square$

**Lemma 1.18.** *The canonical vf-category structure on  $A \subseteq \mathbf{pt}(\mathcal{E})$  is taut. UT: is it always? GS: i hope so! i will write the proof*

*Proof.* We first show that  $\mathbf{Set}$  is taut. Let  $A$  and  $(B_s)_{s:\sigma}$  be sets, we want to show that

$$\mathbf{Set} \left( A, \int_{t:\tau} B_{\phi t} \right)$$

is a sheaf over  $(\tau \xrightarrow{\phi} \sigma) \in \mathbb{F}/\sigma$  for  $J_\infty$ . As  $\mathbf{Set}(A, -)$  is continuous, it is enough to show that

$$\int_{t:\tau} B_{\phi t}$$

is a sheaf over  $(\tau \xrightarrow{\phi} \sigma) \in \mathbb{F}/\sigma$ .

*GS: would be nice to fix on a notation for  $(B, \{B\})$ , i propose  $\delta_B$  but not sure about it*

First, in the case  $\sigma = \star$ , we noticed that  $B^\tau = \mathbb{F}(\tau, (B, \{B\}))$ , so it follows from  $J_\infty$  being sub-canonical.

In the general case, we denote  $\pi: \sum_{s:\sigma} (B_s, \{B_s\}) \rightarrow \sigma$  in  $\mathbb{F}/\sigma$  so that  $\int_{t:\tau} B_{\phi t} = \mathbb{F}/\sigma(\phi, \pi)$ , and so this follows from  $J_\infty$  on  $\mathbb{F}/\sigma$  being subcanonical.

Now we show that  $\mathbf{pt}(\mathcal{E})$  is taut *GS: à continuer*

$\square$

The goal of the following is to prove that the functor  $\mathbf{ev}: \mathcal{E} \rightarrow \mathbf{Sh}(A)$  is an equivalence when  $A$  is a separating set of points of  $\mathcal{E}$ .

We will obtain this by combining two more general results. Note that any vf-functor  $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$  naturally induces a functor  $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$  where, for  $e \in \mathcal{E}$  *GS: i usuallt prefer capital letters for objects in a topos (as their are thought as sheaf)*, the vf-functor  $f^*(e): \mathcal{X} \rightarrow \mathbf{Set}$  is given:

- on objects, by mapping  $x \in \mathcal{X}_0$  to the set  $F(x)(e)$ ;
- on vf-arrows, by mapping  $r: x \multimap_{s:\sigma} y_s$  to the component at  $e$  of the natural transformation  $F(r): x \Rightarrow \prod_{s:\sigma} y_s$ .

Thus, we will now study how properties of  $F$  induce properties on  $f^*$ , in such a way that taking  $F$  to be the inclusion  $A \hookrightarrow \mathbf{pt}(\mathcal{E})$  the corresponding functor  $\mathcal{E} \rightarrow \mathbf{Sh}(A)$  will be an equivalence. Intuitively, we will proceed as if  $f^*$  were the inverse image of a geometric morphism  $\mathbf{Sh}(\mathcal{X}) \rightarrow \mathcal{E}$ , determining sufficient conditions on  $F$  making it localic and conditions making it hyperconnected. Since a geometric morphism which is both localic and hyperconnected is an equivalence, we will derive our conclusion in the case of  $\mathcal{E} \rightarrow \mathbf{Sh}(A)$ .

UT: do we need to assume that  $\mathbf{Sh}(\mathcal{X})$  is a topos, or does infinitary-pretopos suffice? in principle we should have the latter with no problemsGS: I hope we will only need infinitary-pretopos, i guess one can show the accessibility directly, but in our proofs it is deduced from the reconstruction theorem

UT: i assume the next theorems will assume tautness of  $\mathcal{X}$ , or at least make use of tautness of  $\mathbf{pt}(\mathcal{E})$ ?GS: I am actually not sure about how much tautness is used in these theorem, we need to check carefully.

**Theorem 1.19** (Localicness). *If  $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$  is vf-faithful, then any vf-sheaf on  $\mathcal{X}$  is a subquotient of a vf-sheaf in the image of  $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$ .*

**Theorem 1.20** (Sub-hyperconnectedness). *If  $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$  is vf-full, then for each  $e \in \mathcal{E}$  the frame homomorphism  $\mathbf{Sub}_{\mathcal{E}}(e) \rightarrow \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*(e))$  induced by  $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$  is a quotient map.*

## 2. POINTS OF THE LOCALIC REFLECTION (5.3 OF SUJ)

We denote by  $\rho_L: \mathbf{Topos} \rightarrow \mathbf{Loc}$  the usual localic reflection, i.e.  $\rho_L(\mathcal{E}) := \mathbf{Sub}_{\mathcal{E}}(1)$ .

Let  $\mathcal{E}$  be a topos. Note that there is an evident vf-functor  $\mathbf{pt}(\mathcal{E}) \rightarrow \nu_T(\mathbf{pt}(\rho_L(\mathcal{E})))$  defined on objects by mapping  $p \in \mathbf{pt}(\mathcal{E})$  to its restriction  $p|_{\mathbf{Sub}_{\mathcal{E}}(1)}: \mathbf{Sub}_{\mathcal{E}}(1) \rightarrow \mathcal{P}(1)$ , which is a frame homomorphism and hence a point of  $\rho_L(\mathcal{E})$ . By the adjunction  $\rho_T \dashv \nu_T$ , this yields a continuous map  $\Theta: \rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$ .

Identified with its  $\nu_T$ -image, the vf-functor  $\Theta$  is bijective on objects and vf-faithful; we will now show that it is also vf-full, and hence an equivalence of vf-categories.

**Proposition 2.1.** *The functor  $\rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$  is vf-full, i.e., for any  $p$  and  $(q_s)_{s:\sigma}$  such that  $p \preceq_{\sigma} q_s$  in  $\mathbf{pt}(\rho_L(\mathcal{E}))$ , there exists  $p \multimap_{\lambda:\sigma} q_s$  in  $\mathbf{pt}(\mathcal{E})$  for some filter  $\lambda$  inhabited.*

**Lemma 2.2.** *Let  $\mathcal{C}$  be a small filtered category and  $A: \mathcal{C} \rightarrow \mathbf{Set}$  such that there exist a family  $(a_x)_{x:\mathbf{Ob}(\mathcal{C})}$ . Then, there exists an inhabited filter  $\lambda$  such that the reduced product functor  $A^{\lambda}: \mathcal{C} \rightarrow \mathbf{Set}$  defined by  $x \mapsto A(x)^{\lambda}$  admits a global section.*

*Proof.* We consider the filter  $\lambda$  on  $L := \prod_{x:\mathbf{Ob}(\mathcal{C})} \mathbf{Ob}(\mathcal{C}/x)$  defined by:  $L_0 \subseteq L$  is  $\lambda$ -large iff there exists a non-empty finite sub-category  $C \subseteq \mathcal{C}$ , such that  $L_0$  contains all the  $(\theta^x: d^x \rightarrow x)_{x:\mathbf{Ob}(\mathcal{C})} \in L$  with the  $(\theta^x)_{x:C}$  inducing a cone on  $C$ .

It is clear that  $\lambda$  defines a filter, and  $\lambda$  proper is equivalent to  $\mathcal{C}$  being filtered, that is true by hypothesis.

We define a global section of  $A^{\lambda}$  given on  $x \in \mathcal{C}$  by

$$(\theta_*^x(a_{d^x}))_{(\theta^x: d^x \rightarrow x):\lambda} \in A(x)^{\lambda}.$$

We check the functoriality of this family along  $f: y \rightarrow x$ . This amounts to show that

$$(\forall (\theta^x: d^x \rightarrow x): \lambda) f_*((\theta_*^y(a_{d^y}))) = \theta_*^x(a_{d^x})$$



and this is true by construction of  $\lambda$  (as for  $(\theta^x : d^x \rightarrow x) : \lambda$  we can assume  $f \circ \theta^y = \theta^x$ ).  $\square$

*Proof.* Let  $p \in \mathbf{pt}(\rho_T(\mathcal{E}))$ . Fix a small dense full subcategory  $\mathcal{G} \subseteq \mathcal{E}$  of the topos  $\mathcal{E}$  and let  $\mathcal{C}$  be the category of elements of the functor  $(-)_p : \mathcal{G} \rightarrow \mathbf{Set}$ , which is then filtered.

UT: who is  $(-)_p$ ? GS: it is  $p^*$ , like the functor taking the stalk at  $p$ .

The hypothesis  $p \preceq_\sigma(q_s)$  in  $\mathbf{pt}(\rho_L(\mathcal{E}))$  implies that for all  $E \in \mathcal{E}$  such that  $E_p$  is non-empty then  $\int_{s:\sigma} E_{q_s}$  is also non-empty, thus we can apply the above Lemma 2.2 to the functor  $\mathcal{C} \rightarrow \mathbf{Set}$  mapping  $(E, \xi \in E_p) \in \mathcal{C}$  to the set  $\int_{s:\sigma} E_{q_s}$ .

We obtain some filter  $\lambda$  and some  $\alpha_E(\xi) \in \int_{\lambda \cdot \sigma} E_{q_s}$  natural in  $(E, \xi \in E_p) \in \mathcal{C}$ . This gives a natural transformation  $\alpha \in \mathbf{Nat}_{E:S}(E_p, \int_{\lambda \cdot \sigma} E_{q_s})$  that we then can extend to all  $\mathcal{E}$  by density of  $S$ , obtaining thus some vu-arrow  $p \dashv_{\lambda \cdot \sigma}(q_s)$  in  $\mathbf{pt}(\mathcal{E})$  showing that  $p \preceq_\sigma(q_s)$  in  $\rho_T(\mathbf{pt}(\mathcal{E}))$  as desired.  $\square$

**Corollary 2.3.** *The functor  $\rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$  is an equivalence of topological spaces.*

Hence, both ways to take get a topological space from a topos  $\mathcal{E}$ —the points of the localic reflection or the topological reflection of its vf-category of points—coincide.

### 3. PROVING THE THEOREMS

We can deduce

**Theorem 3.1** (5.4 SUJ). *If  $F : \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$  is vf-full, then  $f^* : \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$  is ‘algebraically subhyperconnect’, i.e., for any  $X \in \mathcal{E}$  the map  $\mathbf{Sub}_{\mathcal{E}}(X) \rightarrow \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*X)$  is a quotient of frames (that is, an embedding of locales).*

*Proof.* We consider the pullback

$$\begin{array}{ccc} \mathbf{Et}(f^*X) & \longrightarrow & \mathbf{pt}(\mathcal{E}/X) \cong \mathbf{Et}(X) \\ \text{ét} \downarrow & \lrcorner & \downarrow \text{ét} \\ \mathcal{X} & \xrightarrow{F} & \mathbf{pt}(\mathcal{E}) \end{array}$$

the above map  $\mathbf{Et}(f^*X) \rightarrow \mathbf{pt}(\mathcal{E}/X)$  is still vf-full (vf-fullness is stable by pullback), and that its topological reflection  $\rho_T(\mathbf{Et}(f^*X)) \rightarrow \rho_T(\mathbf{pt}(\mathcal{E}/X))$  also (vf-fullness is stable by topological reflection). Then by the above lemma we get

$$\rho_T(\mathbf{Et}(f^*X)) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}/X))$$

is vf-full. But a vf-full functor between vf-posets induces a quotient on the frames of opens, so

$$\mathcal{O}(\mathbf{pt}(\rho_L(\mathcal{E}/X))) \rightarrow \mathcal{O}(\rho_T(\mathbf{Et}(f^*X))) \cong \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*X)$$

is surjective. It remains to notice that  $\mathbf{Sub}_{\mathcal{E}}(X) \rightarrow \mathcal{O}(\mathbf{pt}(\rho_L(\mathcal{E}/X)))$  is surjective, and this follows from the more general fact that  $\mathbf{Sub}(L) \rightarrow \mathbf{Sub}(\mathcal{O}(\mathbf{pt}(L)))$  is surjective, i.e., the restriction of a local to its points is a sublocal.  $\square$

**Corollary 3.2.** *The functor  $\mathcal{E} \rightarrow \mathbf{Sh}(\mathbf{pt}(\mathcal{E}))$  induces equivalences on subobjects.*

*Proof.* Its surjective on subobjects by the above. Using that  $\mathcal{E}$  has enough points we conclude it is injective.  $\square$

Now we can show the second part.

**Theorem 3.3.** *If  $F : \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$  is faithful, then  $f^* : \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$  is ‘algebraically localic’, i.e., any  $A \in \mathbf{Sh}(\mathcal{X})$  is a subquotient of a vf-sheaf  $f^*X$  for some  $X \in \mathcal{E}$ .*

**Lemma 3.4.** *Let  $\pi : \mathcal{A} \rightarrow \mathcal{X}$  étale with  $\pi : a \mapsto p$ , then the following are equivalent:*

- (i) *for any two parallel vf-arrows  $f, g : p \multimap (-)$ ,  $f_*a = g_*a$ .*
- (ii) *for any vf-arrow  $f : p \multimap_\sigma q_s$  with lift  $\tilde{f} : a \multimap_\sigma b_s$ , then  $\sigma \rightarrow \mathcal{A}_0$  factorizes through  $\sigma \rightarrow q_*(\sigma)$ .*
- (iii) *for any vf-arrow  $f : p \multimap_\sigma q_s$  with lift  $\tilde{f} : a \multimap_\sigma b_s$ , then  $\pi$  is injective on  $b_*(\sigma)$  and so establish an isomorphism  $b_*\sigma \cong q_*\sigma$ .*
- (iv) *if  $(b_s)_\sigma$  converges to  $a$ , then  $\pi$  is  $b_*\sigma$ -injective.*
- (v) *there exists an open  $a \in V \subseteq \mathcal{A}$  such that  $V \rightarrow \mathcal{X}$  is an embedding.*

*SUJ 4.7(i  $\Rightarrow$  ii)* we note that  $(b_s) \in \mathcal{A}_0^\sigma$  can be lift to some element of  $\mathcal{A}_0^{q_*\sigma}$  using the sheaf condition on  $\sigma \mapsto \mathcal{A}_0^\sigma$ .

- (ii  $\Rightarrow$  iii) **GS: really constructive?** denoting  $\phi : \sigma \rightarrow q_*\sigma$  and  $\tau = \phi_*\sigma$  we have some  $(q'_t)_{t:\tau}$  such that  $q'_{\phi s} = q_s$ , and similarly, some  $(b'_t)_{t:\tau}$  such that  $b'_{\phi s} = b_s$ . Moreover, the families  $(b'_t)$  and  $(q'_t)$  are  $\tau$ -injective.
- (iii  $\Rightarrow$  iv) if  $(b_s)_\sigma$  converges to  $a$ , by the proposition above we have a vf-arrow  $a \multimap_\sigma (b_s)$ , and we can apply (iii) on its image by  $\pi$ .
- (iv  $\Rightarrow$  v) apply (iv) to the filter of open neighborhoods of  $a$ .
- (v  $\Rightarrow$  i) clear.

□

*proof of the theorem – SUJ 5.5-5.6.* Let  $A \in \text{Sh}(\mathcal{X})$ , it is enough to show that for any  $a \in A_p$  there is  $X \in \mathcal{E}$ ,  $V \subseteq f^*(X)$ , and a map  $V \rightarrow A$  touching  $a$ . It is enough to show that we can find a  $X \in \mathcal{E}$  and  $\xi \in X_{Fp}$  such that

$$\forall f, g : p \multimap_\sigma q_s \quad F(f)_*\xi = F(g)_*\xi \Rightarrow f_*a = g_*a.$$

as then it is easy to conclude by applying the above on the map  $(a, \xi) \in \mathcal{A} \times_{\mathcal{X}} \text{Et}(f^*X) \rightarrow \text{Et}(f^*X)$ . We now show the existence of such a  $\xi \in X_{Fp}$ .

On prend  $S$  un (petit) ensemble de  $s = (f_s, g_s)$  couple de  $\tau_s$ -flèches parallèles de domaine  $p$ , tq pour toute paire  $(f, g)$  de  $\lambda$ -flèches parallèles de domaine  $p$  il existe épaississement  $\phi : \lambda \rightarrow \tau$  et  $(f', g') \in S$  tels que  $f = \phi^*f'$  et  $g = \phi^*g'$ .

On définit un filtre  $\sigma$  sur  $S$ . Pour tout  $\xi \in X_{Fp}$  on note  $A_\xi = \{(f, g) \in S \mid F(f)_*\xi = F(g)_*\xi\}$ , et pour  $\xi \in X_{Fp}$  et  $\zeta \in Y_{Fp}$  on a  $(\xi, \zeta) \in (X \times Y)_{Fp}$ , et  $A_\xi \cap A_\zeta = A_{(\xi, \zeta)}$ . Donc les  $(A_\xi)$  définissent un filtre  $\sigma$  sur  $S$  (une partie de  $S$  est  $\sigma$ -large ssi elle contient un  $A_\xi$ ).

On pose alors  $f = (f_s)_{s:\sigma}$  et  $g = (g_s)_{s:\sigma}$ , ce sont deux  $(s : \sigma) \cdot \tau_s$ -flèches. De plus, par définition de  $\sigma$ , pour tout  $\xi \in X_p$ ,  $F(f)_*\xi = F(g)_*\xi$ , donc  $F(f) = F(g)$ , et par fidélité de  $F$  on déduit que  $f = g$ .

Ainsi,  $f_*a = g_*a$ , et donc  $(f_s)_*a = (g_s)_*a$  pour tout  $s : \sigma$ , i.e., il existe un  $\xi \in X$  tel que pour tout  $s \in A_\xi$ ,  $(f_s)_*a = (g_s)_*a$ , i.e., tel que pour tout  $(f, g) \in S$ , si  $F(f)_*\xi = F(g)_*\xi$  alors  $f_*a = g_*a$ .

Maintenant, pour  $(f, g)$  deux  $\lambda$ -flèches parallèles sortant de  $p$  quelconque, on peut par construction de  $S$  trouver un  $\phi : \lambda \rightarrow \tau$  couvrant et  $(f', g') \in S$  tels que  $f = \phi^*f'$  et  $g = \phi^*g'$ . Alors si  $F(f)_*\xi = F(g)_*\xi$ ,  $F(\phi^*f')_*\xi = F(\phi^*g')_*\xi$ , et  $\phi^*F(f')_*\xi = \phi^*F(g')_*\xi$ , mais comme  $\phi$  est couvrant,  $\phi^*$  est injectif et donc  $F(f')_*\xi = F(g')_*\xi$ , d'où  $f'_*a = g'_*a$  et  $\phi^*f'_*a = \phi^*g'_*a$  soit  $f_*a = g_*a$ . □

**Corollary 3.5.** *The functor  $\mathcal{E} \rightarrow \text{Sh}(\text{pt}(\mathcal{E}))$  is an equivalence of categories.*

*Proof.* todo

□



## APPENDIX A. TAUTNESS: THE SHEAF DEFINITION

We first begin with the most concrete and most straightforward definition of tautness.

First notice that if  $f = \phi^*g : p \multimap_\tau (q_{\phi t})_{t:\tau}$  is the thickening of some  $g : p \multimap_\sigma (q_s)$ , then for any  $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$  equalizing  $\phi$  (i.e.,  $\phi \circ \psi_1 = \phi \circ \psi_2$ ) we have  $\psi_1^*f = \psi_2^*f$ .

A vu-category  $\mathcal{X}$  is said *taut* if the reverse is true, we can always unthicken a vu-arrow if we can (and if so, in a unique way):

**Definition A.1.** For any  $f : p \multimap_\tau (q_{\phi t})_{t:\tau}$ , such that

$$(\forall \psi_1, \psi_2 : \lambda \rightrightarrows \tau \text{ equalizing } \phi) \psi_1^*f = \psi_2^*f$$

(we will say  $f$  is unthickenable along  $\phi$  if its the case)

there exists a unique  $g : p \multimap_\sigma (q_s)$  such that  $f = \phi^*g$ .

This definition can be understand as a ‘internal (Cauchy-)completeness of the objects of  $\mathcal{X}$ ’: in the case  $\mathcal{X}$  being the ultracategory of complete metric spaces, it really gives that any Cauchy-ultrasequence converge.

**Definition A.2.** Another way to express the definition above is to ask the following functor  $(\mathbb{U}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$  whose functoriality encodes the thickening operation

$$(\phi : \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t:\tau})$$

to be a sheaf for the atomic topology  $\mathbf{Sh}(\mathbb{U}_{at})$ .

This definition can thus be understood as ‘the thickening is a sheaf for the topology on  $\mathbb{U}$ ’,

**Remark A.3.** In light of Eliasson result that  $\mathbf{Sh}(\mathbb{U}_{at}) = \mathbf{Sh}(\mathbb{F}, J_\infty)$  we expect that, under tautness assumption, we can identify vu and vf categories. Thus suggesting that we get a notion of 1-space that is no more dependent of the ultrafilter calculus.

## APPENDIX B. VU-VF?

Let  $\sigma \in \mathcal{F}$  be a filter, we denote  $\mathbf{Ext}(\sigma) := \{\dot{\sigma} \hookrightarrow \sigma\}$  the set of the subobjects of  $\sigma$  in  $\mathcal{F}$  that are ultrafilters. Moreover  $\mathbf{Ext}(\sigma) \subseteq \beta(S)$  has a compact Hausdorff topology (and this topology doesn’t depend of the choice of  $S$ ), for  $(\dot{\sigma}_\ell)_{\ell:\lambda}$  in  $\mathbf{Ext}(\sigma)$ , we denote  $\int_{\ell:\lambda} \dot{\sigma}_\ell \in \mathbf{Ext}(\sigma)$  its limit in  $\mathbf{Ext}(\sigma)$ .

**Remark B.1.** For a choice of underlying set  $S$ , we can identify  $\mathbf{Ext}(\sigma)$  with the set of ultrafilters  $\dot{\sigma} \in \beta(S)$  extending  $\sigma$ . Moreover  $(\dot{\sigma} \hookrightarrow \sigma)_{\dot{\sigma} \in \mathbf{Ext}(\sigma)}$  is jointly epic in  $\mathbb{F}$ , and so is a cover for  $J_\infty$ .

Moreover, the category  $\mathbb{U}/\sigma$  is isomorphic to the disjoint union  $\bigsqcup_{\dot{\sigma} \in \mathbf{Ext}(\sigma)} \mathbb{U}/\dot{\sigma}$ .

**Construction B.2.** Let  $\mathcal{X}$  be an vu-category, we want to define a vf-category  $\mathbf{vf}(\mathcal{X})$ . The objects are the same. For  $p, (q_s)_{s:\sigma}$  and  $\sigma \in \mathbb{F}$ , we define

$$\mathbf{vf}(\mathcal{X})(p, (q_s)_{s:\sigma}) \subseteq \prod_{\dot{\sigma} \in \mathbf{Ext}(\sigma)} \mathcal{X}(p, (q_s)_{s:\dot{\sigma}})$$

to be the subset consisting of families  $(f_{\dot{\sigma}} : p \multimap_{\dot{\sigma}} q_s)$  satisfying the following *compatibility condition*:

for any (injective) family  $(\dot{\sigma}_\ell \in \mathbf{Ext}(\sigma))_{\ell:\lambda}$ , the vu-arrow

$$(f_{\dot{\sigma}_\ell}) \circ \delta : p \multimap_\lambda (p \multimap_{\dot{\sigma}_\ell} q_{t\ell})$$

is equal the thickening of  $f_{\int_{\ell:\lambda} \dot{\sigma}_\ell}$  along  $\sum_{\ell:\lambda} \dot{\sigma}_\ell \rightarrow \int_{\ell:\lambda} \dot{\sigma}_\ell$ .

**Remark B.3.** As  $\mathbf{Ext}(\sigma) = \sigma$  if  $\sigma$  is an ultrafilter,  $\mathbf{vf}(\mathcal{X})$  has the same vu-arrows that  $\mathcal{X}$ .

**Construction B.4.** We now to define the composition. Let  $f : p \multimap_{\sigma} q_s$  in  $\mathbf{vf}(\mathcal{X})(p, (q_s)_{s:\sigma})$  given by  $f = (f_{\dot{\sigma}})_{\dot{\sigma}:\text{Ext}(\sigma)}$ , and a  $\sigma$ -family of  $(g^s : q_s \multimap_{t:\tau_s} r^s_t)$  in  $\mathbf{vf}(\mathcal{X})(q_s, (r^s_t)_{t:\tau_s})$  given by  $g^s = (g^s_{\dot{\tau}_s})_{\dot{\tau}_s:\text{Ext}(\tau_s)}$ .

To define the composite, we need for each  $\theta \in \text{Ext}(\sum_{s:\sigma} \tau_s)$  some

$$h_{\theta} : p \multimap_{(s,t):\lambda} r^s_t$$

for any  $\dot{\sigma} \in \text{Ext}(\sigma)$  and  $(\dot{\tau}_s \in \text{Ext}(\tau_s))_{s:\dot{\sigma}}$ , we let

$$h_{\sum_{\dot{\sigma}} \dot{\tau}_s} = (g^s_{\dot{\tau}_s}) \circ f_{\dot{\sigma}}$$

prescribing the value of  $h$  on the subset

$$\sum_{\dot{\sigma} \in \text{Ext}(\sigma), (\dot{\tau}_s \in \text{Ext}(\tau_s))} \int_{s:\sigma} \text{Ext}(\dot{\tau}_s) \subseteq \text{Ext} \left( \sum_{s:\sigma} \tau_s \right)$$

that is dense (any point in the right space is a limit point of an ultrafamily from the left space).

Hence the value of  $h$  is determined everywhere, we need to show the compatibility condition for it...

**Lemma B.5.** *Let  $\lambda$  be an ultrafilter, and  $\sigma_{\ell} \in \text{Ext}(F)$  converging to  $\int_{\ell:\lambda} \sigma_{\ell} = \sigma \in \text{Ext}(F)$ , and for  $s : \sigma_{\ell}$  an ultrafilter  $\tau_{\ell}^s \in \text{Ext}(G_s)$  converging to  $\int_{\ell:\lambda} \tau_{\ell}^s = \tau^s \in \text{Ext}(G_s)$ , , then*

$$(h_{\sum_{s:\sigma_{\ell}} \tau_{\ell}^s}) \circ \delta : a \rightarrow (a)_{\ell:\lambda} \rightarrow (c_{s,t})_{(\ell:\lambda) \cdot (s:\sigma_{\ell}) \cdot (t:\tau_{\ell}^s)}$$

and

$$(h_{\sum_{s:\sigma} \tau^s}) : a \rightarrow (c_{s,t})_{(s:\sigma) \cdot (t:\tau^s)}$$

coincide (modulo thickening).

## APPENDIX C. DIRECT IMAGE

We try to give an explicit description of the direct image.

We take  $F : \mathcal{X} \rightarrow \mathcal{Y}$  a vf-functor, some  $A \in \mathbf{Sh}(\mathcal{X})$ , and want to define  $B = f_*(A) \in \mathbf{Sh}(\mathcal{Y})$  the direct image (this always exists by the special adjoint theorem if  $A$  is bounded) that can be categorically thought as a right Kan-extension.

We define  $B(y)$  as the data of families  $(\xi^h_s \in A(x_s))_{s:\sigma}$  for all  $h : y \multimap_{\sigma} F(x_s)$ , such that

- (i) if  $h : y \multimap_{\sigma} F(x_s)$  and  $(f_s : x_s \rightarrow x'_s)$ , then  $\int A(x_s) \rightarrow \int A(x'_s)$  sends  $\xi^h$  to  $\xi^{(f_s) \circ h}$ .
- (ii) if  $h : y \multimap_{\sigma} F(x_s)$  and  $\phi : \tau \rightarrow \sigma$ , then  $\int_{\sigma} A(x_s) \rightarrow \int_{\tau} A(x_{\phi t})$  sends  $\xi^h$  to  $\phi^*(\xi^h)$ .
- (iii) if  $(h^{\ell} : y \multimap_{\sigma_{\ell}} F(x_s^{\ell}))_{\ell:\lambda}$  merge into  $h : y \multimap_{\lambda} (y \multimap_{\sigma_{\ell}} F(x_s^{\ell}))$  then  $\xi^h \in \int_{\ell:\lambda} \int_{s:\sigma_{\ell}} A(x_s^{\ell})$  equals  $(\xi^{h^{\ell}} \in \int_{s:\sigma} A(x_s^{\ell}))_{\ell:\lambda}$

This can be seen as taking the rep-vf-category of vf-arrows  $y \multimap_{-} F(-)$ , given a vf-sheaf on it given by  $\int_{-} A(-)$ , and saying  $B(y)$  is the global element of this sheaf.

We need to show this is small, and functorial. The smallness can follow from assuming  $\mathcal{X}$  and  $\mathcal{Y}$  small 1-spaces (but we can surely ask less). Let's try to give the functoriality, we take  $g : y \multimap_{\tau} y'_t$  in  $\mathcal{Y}$ , and some  $(\xi^h)_h$  defining an element of  $B(y)$ , we want an element of  $\int_{t:\tau} B(y'_t)$ .

We have an inclusion (where the limits are taking with the condition above, maybe not exactly limits...)

$$\int_{t:\tau} B(y'_t) = \int_{t:\tau} \lim_{h:y'_t \multimap_{\sigma_t} F(x_s^t)} \int_{s:\sigma_t} A(x_s^t) \hookrightarrow \lim_{(h_t:y'_t \multimap_{\sigma_t} F(x_s^t))} \int_{t:\tau} \int_{s:\sigma_t} A(x_s^t)$$

The  $(\xi^h)_h$  gives for any  $(h_t : y'_t \multimap_{\sigma_t} F(x_s^t))_{t:\tau}$  an element  $(\xi^{(h_t) \circ g})$  of  $\int_{\tau} \int_{\sigma_t} A(x_s^t)$ , so an element of the right set. We need to show it is in the left one.

## APPENDIX D. TAUTNESS 2: THE STACK DEFINITION

In the posetal case, small vu-poset are sober if and only if they are taut, we denote  $X \rightarrow \rho_T(X)$  the tautification of a vu-poset. One can extend to the following

**Definition D.1.** A vu-category  $\mathcal{X}$  if it is right orthogonal to all  $X \rightarrow \rho_T(X)$  for (small)  $X$  vu-poset, i.e., precomposing with  $X \rightarrow \rho_T(X)$  induces an equivalence of category

$$\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \xrightarrow{\sim} \mathbf{vuCat}(X, \mathcal{X}).$$

**Lemma D.2.** Let  $f : p \multimap_\tau (q_{\phi t})$ . We chose  $\phi = T \rightarrow S$  representing  $\tau \rightarrow \sigma$ , and consider the vu-poset  $X = \{x\} \sqcup S$  with only convergences  $x \preceq_\lambda (\rho_\ell)$  for some  $\rho \in S^\lambda$  with  $\rho : \lambda \rightarrow \sigma$  iff  $\rho$  factorizes (not necessarily uniquely) through  $\rho : \lambda \rightarrow \tau \rightarrow \sigma$ .

Then  $f$  is unthickenable, iff the map  $X \rightarrow \mathcal{X}$  mapping  $x \preceq_\tau (t)$  to  $f$ , extends to a vu-functor.

*Proof.* We suppose it is unthickenable. We then have a functor  $X \rightarrow \mathcal{X}$  sending  $x \mapsto p$  and  $s \mapsto q_s$ , and  $x \preceq_\lambda (\rho_\ell)$  is sent on  $\rho'^* f : p \multimap_\lambda (q_{\rho t})$ , (for any  $\rho' : \lambda \rightarrow \tau$  factorizing  $\rho$ , it doesn't depend of the choice of it by hypothesis).

The other direction is similar. □

*proof: equivalence with the sheaf definition.* Let suppose the orthogonality condition, and  $f : p \multimap_\tau (q_{\phi t})$  such that  $\psi_1^* f = \phi_2^* f$  for all  $\phi_i$  equalizing  $\phi$ . And we take  $X$  as in the lemma above.

Then  $\rho_T(X)$  is given by the topological space  $X = \{x\} \sqcup S$  with only convergence  $x \preceq_\sigma (s)$ .

Hence, the orthogonality gives us a unique  $g : p \multimap_\sigma (q_s)_{s:\sigma}$  unthickening  $f$  as wanted.

We suppose now the sheaf condition. The uniqueness of the unthickening gives that  $\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \rightarrow \mathbf{vuCat}(X, \mathcal{X})$  is fully-faithful:

indeed, let  $F, G : \rho_T(X) \rightarrow \mathcal{X}$  and a natural transformation on  $X$ , i.e., for any  $x \in X$  a  $\alpha_x : F(x) \rightarrow G(x)$  such that for any  $f : x \preceq_\sigma (y_s)$  in  $X$  we have  $G(f) \circ \alpha_x = (\alpha_{y_s}) \circ F(f)$ . Then it is clear that this can be extended in at most one way to a natural transformation  $\alpha : F \Rightarrow G$ , and that this is the case iff the naturality condition holds for any  $f$  in the tautification of  $X$ . But if  $f : x \preceq_\sigma (y_s)$  is in the tautification of  $X$ , there exists a  $\phi$  such that  $\phi^* f : x \preceq_\tau (y_{\phi t})$  is in  $X$  and so  $G(\phi^* f) \circ \alpha_x = (\alpha_{y_{\phi t}}) \circ F(\phi^* f)$ , that means  $\phi^* G(f) \circ \alpha_x = (\alpha_{y_{\phi t}}) \circ \phi^* F(f)$ , or again  $\phi^*(G(f) \circ \alpha_x) = \phi^*((\alpha_{y_s}) \circ F(f))$ , and we conclude  $G(f) \circ \alpha_x = (\alpha_{y_s}) \circ F(f)$  by uniqueness of the unthickening.

We now show  $\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \rightarrow \mathbf{vuCat}(X, \mathcal{X})$  to be essentially surjective, we take some  $F : X \rightarrow \mathcal{X}$  and we want to show it can be extended to some  $G : \rho_T(X) \rightarrow \mathcal{X}$ . On the objects  $G$  is  $F$ , we take now a vu-arrow  $f : x \preceq_\sigma (y_s)$  in  $\rho_T(X)$ , i.e., such that  $\phi^* f : x \preceq_\tau (y_{\phi t})$  is true in  $X$ . We can show that  $F(\phi^* f)$  satisfies the condition to be unthicken (we can take the  $X_0 = \{x\} \sqcup S$  from above and sent it to  $X$  and use the lemma above). Hence  $G$  is defined on vu-arrows, the functoriality come again by the functoriality of  $F$  by uniqueness of the unthickening. □

## APPENDIX E. TAUTNESS 3: THE ALGEBRAIC DEFINITION

Let  $\phi : \tau \rightarrow \nu$  and  $f : p \multimap_\tau (q_{\phi t})$  unthickenable. The magic choice lemma on  $\mathbf{U}$  allows us to 'make  $\phi$  a projection': we can find  $\lambda$  and  $\psi : \lambda \cdot \sigma \rightarrow \tau$  such that  $\phi \circ \psi = \pi_1$ . Then both maps  $\lambda \cdot \lambda \cdot \sigma \rightrightarrows \lambda \cdot \sigma \rightarrow \tau$  equalize  $\phi$ , and so  $\psi^* f : p \multimap_{\lambda \cdot \sigma} (q_s)$  have the same thickening along both projections  $\lambda \cdot \lambda \cdot \sigma$ .

Hence we get the following

**Proposition E.1.** *For  $\mathcal{X}$  to be taut, it is enough to ask for any  $f : p \multimap_{\lambda \cdot \sigma} (q_s)$  such that  $\pi_1^* f = \pi_2 * f : p \multimap_{\lambda \cdot \lambda \cdot \sigma} (q_s)$  to be of the form  $f = !_{\lambda}^* g$  for some  $g : p \multimap_{\sigma} (q_s)$ .*

Or, abstracting the hom functor

**Proposition E.2.** *For  $H : (\mathbb{U}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$  to be a sheaf, it is enough to have for any  $\phi : \tau \rightarrow \sigma$  and  $\lambda \in \mathbb{U}$ , the coequalize in  $\mathbb{U}$*

$$\lambda \cdot \lambda \cdot \tau \rightrightarrows \lambda \cdot \tau \rightarrow \tau$$

*induces the coequalizer in  $\mathbb{U}/\sigma$*

$$\lambda \cdot \lambda \cdot \phi \rightrightarrows \lambda \cdot \phi \rightarrow \phi$$

*induces a equalizer of sets*

$$A(\lambda \cdot \lambda \cdot \phi) \rightrightarrows A(\lambda \cdot \phi) \rightarrow A(\phi)$$

It is not hard to check that this à priori stronger condition is satisfied on the sober vu-category. We will show it is actually not stronger.

On peut montrer? qu'un sheaf sur  $\mathbb{U}/\sigma$  preserve les coeq de la forme

$$(k : \kappa) \cdot \lambda_k \cdot \lambda_k \cdot \tau_k \rightrightarrows (k : \kappa) \cdot \lambda_k \cdot \tau_s \rightarrow (k : \kappa) \cdot \tau_k$$

pour une certaine map  $(k : \kappa) \cdot \tau_k \rightarrow \sigma$

**Proposition E.3.** *Let  $H : \mathbb{U}^{\text{op}} \rightarrow \mathbf{Set}$  a sheaf, then*

$$H(\star) \rightarrow H(\tau) \rightrightarrows H(\tau \cdot \tau)$$

*is an equalizer.*

*Proof.* Let  $\xi \in H(\tau)$  such that  $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\tau \cdot \tau)$ , for  $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$  we will show that  $\psi_1^*(\xi) = \psi_2^*(\xi)$ :

$$\begin{array}{ccc} \lambda \cdot \lambda \rightrightarrows_{\pi_2}^{\pi_1} \lambda & & \\ \psi_1 \cdot \psi_2 \downarrow & & \psi_1 \downarrow \downarrow \psi_2 \\ \tau \cdot \tau \rightrightarrows_{\pi_2}^{\pi_1} \tau & & \end{array}$$

As  $H$  sends any map to a mono, it is enough to show that  $\pi_1^*(\psi_1^*(\xi)) = \pi_1^*(\psi_2^*(\xi))$

$$\pi_1^*(\psi_1^*(\xi)) = (\psi_1 \cdot \psi_2)^*(\pi_1^*(\xi)) = (\psi_1 \cdot \psi_2)^*(\pi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$$

Hence, for all  $\psi_1, \psi_2 : \tau \rightrightarrows \sigma$  we have  $\pi_1^*(\psi_1^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$

In particular (using  $\psi_2 \cdot \psi_2$  instead of  $\psi_1 \cdot \psi_2$ ) we get

$$\pi_1^*(\psi_2^*(\xi)) = (\psi_2 \cdot \psi_2)^*(\pi_1^*(\xi)) = (\psi_2 \cdot \psi_2)^*(\pi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$$

$\pi_1^*(\psi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$ . and so  $\pi_1^*(\psi_1^*(\xi)) = \pi_2^*(\psi_2^*(\xi)) = \pi_1^*(\psi_2^*(\xi))$  as wanted.  $\square$

A first generalization

**Proposition E.4.** *Let  $H : \mathbb{U}^{\text{op}} \rightarrow \mathbf{Set}$  a sheaf, then*

$$H(\kappa) \rightarrow H(\tau \cdot \kappa) \rightrightarrows H(\tau \cdot \tau \cdot \kappa)$$

*is an equalizer.*

*Proof.* Let  $\xi \in H(\tau \cdot \kappa)$  such that  $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\tau \cdot \tau \cdot \kappa)$ , for equalizing

$$\psi_1, \psi_2 : \lambda \rightrightarrows \tau \cdot \kappa \rightarrow \kappa$$

we will show that  $\psi_1^*(\xi) = \psi_2^*(\xi)$ :

It is enough to give some

$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \tau \cdot \tau \cdot \kappa & \xrightarrow[\pi_2]{\pi_1} & \tau \cdot \kappa \xrightarrow{\pi_2} \kappa \end{array}$$

such that  $\pi_2 f_1 = \pi_2 f_2$  and  $\pi_1 f_1 = \psi_1 g$ ,  $\pi_1 f_2 = \psi_2 g$  and we conclude as upside.  $\square$

**Proposition E.5.** *Let  $H : \mathbb{U}^{\text{op}} \rightarrow \text{Set}$  a sheaf, then*

$$H(\sigma \cdot \kappa_s) \rightarrow H(\sigma \cdot \tau_s \cdot \kappa_s) \rightrightarrows H(\sigma \cdot \tau_s \cdot \tau_s \cdot \kappa_s)$$

*is an equalizer.*

*Proof.* Let  $\xi \in H(\sigma \cdot \tau_s \cdot \kappa_s)$  such that  $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\sigma \cdot \tau_s \cdot \tau_s \cdot \kappa_s)$ , for equalizing

$$\psi_1, \psi_2 : \lambda \rightrightarrows \sigma \cdot \tau_s \cdot \kappa_s \rightarrow \sigma \cdot \kappa_s$$

we will show that  $\psi_1^*(\xi) = \psi_2^*(\xi)$ :

It is enough to give some

$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \sigma \cdot \tau_s \cdot \tau_s \cdot \kappa & \xrightarrow[\pi_2]{\pi_1} & \sigma \cdot \tau_s \cdot \kappa_s \xrightarrow{\pi_2} \sigma \cdot \kappa_s \end{array}$$

such that  $\pi_2 f_1 = \pi_2 f_2$  and  $\pi_1 f_1 = \psi_1 g$ ,  $\pi_1 f_2 = \psi_2 g$  and we conclude as upside.  $\square$

**Question E.6.** En générale, quelle condition il faut sur un coeq

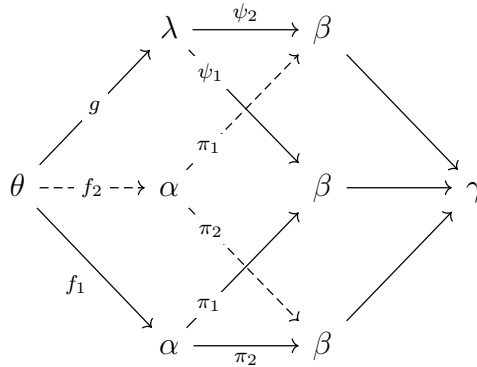
$$\alpha \rightrightarrows \beta \rightarrow \gamma$$

pour avoir:

pour tout  $\psi_1, \psi_1 : \lambda \rightrightarrows \beta$  equalizing  $\beta \rightarrow \gamma$  there exists such a diagram

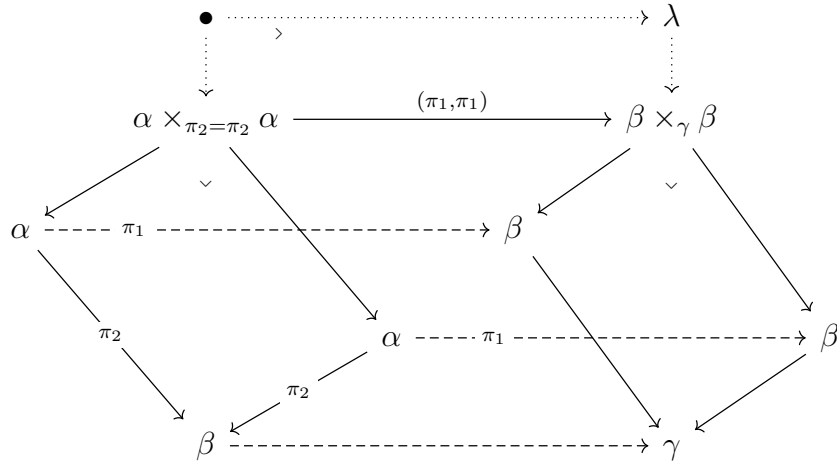
$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \alpha & \xrightarrow[\pi_2]{\pi_1} & \beta \longrightarrow \gamma \end{array}$$

such that  $\pi_2 f_1 = \pi_2 f_2$  and  $\pi_1 f_1 = \psi_1 g$ ,  $\pi_1 f_2 = \psi_2 g$ .



In  $\mathbb{U}$  we can complete span, but not in general cubes! (it is not filtered category as we cannot complete parallel arrow, a demi-cubes has equations in it contrary to a demi-square).

This is equivalent to ask for



for any  $\lambda \rightarrow \beta \times \beta$  (with  $\lambda$  ultrafilter), the pullback along  $\alpha \times_{\pi_2=\pi_2} \alpha \rightarrow \beta \times_{\gamma} \beta$  gives a proper filter. In particular, it is enough (and also necessary i think) to ask for

$$\alpha \times_{\pi_2=\pi_2} \alpha \longrightarrow \beta \times_{\gamma} \beta$$

to be covering!

We thus need to show that the following are covering

$$(\tau, \tau) \times_{\star, \tau} (\tau, \tau) \xrightarrow{\pi_1, \pi_1} \tau \times \tau$$

$$(\tau, \tau, \kappa) \times_{\star, \tau, \kappa} (\tau, \tau, \kappa) \xrightarrow{\pi_1, \pi_1} (\tau, \kappa) \times_{\kappa} (\tau, \kappa)$$

$$(\sigma, \tau_s, \tau_s, \kappa_s) \times_{\sigma, \star, \tau_s, \kappa_s} (\sigma, \tau_s, \tau_s, \kappa_s) \xrightarrow{\pi_1, \pi_1} (\sigma, \tau_s, \kappa_s) \times_{\sigma, \kappa_s} (\sigma, \tau_s, \kappa_s)$$

and these morphisms are respectively isomorphic to

$$(\tau \times \tau), \tau \longrightarrow \tau \times \tau$$

$$(\tau \times \tau), \tau, \kappa \longrightarrow (\tau \times \tau), \kappa$$

$$\sigma, (\tau_s \times \tau_s), \tau_s, \kappa_s \longrightarrow \sigma, (\tau_s \times \tau_s), \kappa_s$$

## APPENDIX F. FILTERS AS GERMS OF CLOSED, AN ALTERNATIVE

We recall that an ultrafilter  $\sigma \in \mathbb{U}$  can be represented as the germ of  $\sigma \in \beta(S)$ . Concretely, the category of ultrafilters can be seen as a (not-full) subcategory of the category of germs of topological spaces (see [Moer]). Or exactly of as the germ of the (non-full) subcategory of topological spaces given by  $\beta(S)$  and  $\beta(f)$ .

We give now a similar realization of filters, that differs from the one of [Moer]. A filter  $\sigma \in \mathbb{F}$  on  $S$ , can be identified to the subset of the  $\sigma \in \beta(S)$  that extend  $\sigma$ , this is a closed subset that we denoted  $\text{Ext}(\sigma) \subseteq \beta(S)$ . And we can do the same than above, replacing germ by open neighborhood of a closed subset.

Note that if one take  $\text{Ext}(\sigma) \subseteq \beta(S)$ , and quotient  $\text{Ext}(\sigma)$  onto one point, then we get the germ described in [Moer].

This equivalent definition of  $\mathbb{F}$  can be then given as follows (but its (à priori?) non-constructive! still an interesting intuition though). The objects are pairs of a set  $S$  with a closed subspace  $K \subseteq \beta(S)$  (this corresponds bijectively to the filter whose  $S_0 \subseteq S$  is large iff  $K \subseteq \beta(S_0)$ )

**Proposition F.1.** *These constructions are inverse from each other.*



*Proof.* From  $\sigma$  a filter on  $S$ , we get  $K \subseteq \beta(S)$  of ultrafilter extending  $\sigma$ , we need to show that for  $S_0 \subseteq S$ ,  $S_0 \in \sigma$  iff  $K \subseteq \beta(S_0)$ : if  $S_0 \in \sigma$  and  $\sigma \in K$ , i.e.,  $\sigma$  extending  $\sigma$ , then  $S_0$  is also  $\sigma$ -large and so  $\beta(S_0) \ni \sigma$ ; conversely if  $S_0 \notin \sigma$  then we can extend  $\sigma$  into an ultrafilter  $\sigma'$  containing  $S \setminus S_0$  and so  $\sigma' \in K$  but  $\sigma' \notin \beta(S_0)$ .

From  $K \subseteq \beta(S)$  a closed subset, we get  $\sigma$  the filter of  $S_0$  such that  $K \subseteq \beta(S_0)$ , and we need to show for  $\sigma \in \beta(S)$ , that  $\sigma \in K$  iff it extends  $\sigma$ : if  $\sigma \in K$  and  $S_0 \in \sigma$ , i.e.,  $K \subseteq \beta(S_0)$ , then  $\sigma \in \beta(S_0)$  and  $S_0 \in \sigma$  as wanted; conversely if  $\sigma \notin K$  then, as  $K$  is closed and the topology is generated by the  $\beta(S_0)$ , we can find  $S_0$  such that  $K \subseteq \beta(S_0)$  but  $\sigma \notin \beta(S_0)$ .  $\square$

All of this uses ultrafilter... but we can go localic!

The local  $\beta(S)$  is given by the poset  $Fil(S)$  of filters on  $S$  ordered by usual inclusion (as above the open  $U \subseteq \beta(S)$  corresponds to the filter of  $S_0$  such that  $\beta(S_0) \cup U = \beta(S)$ ), that is indeed a frame.

We can defined the category of germs of closed subset: objects are given by a locale  $L$  and an open  $U \in \mathcal{O}(L)$ , and a map from  $(L, U)$  to  $(L', U')$  is given by an open  $V \in \mathcal{O}(L)$  such that  $V \vee U = L$  and a map of locales  $f : V \rightarrow L'$  such that  $f^{-1}(U') \subseteq U \wedge V$ , modulo the equivalence relation identifying two such  $(V, f)$   $(V', f')$  if  $f$  and  $f'$  coincide when restricted to  $V \wedge V'$ .

This defined the category of germes of closed subsets. If we restricted to locales of the form  $\beta(S)$  (or explicetely the frame given by the poset of filters on  $S$ ) and ask  $f$  to be open maps, we get the category of filter. It would be interesting to see what happens if we relax the condition: considering all locales but still open maps, or all locals and all maps.

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